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Randomized variable-metric monotone operator learning for linear inverse problems

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Background: Monotone Operator Learning (MOL) [1]

- **Problem**: Recover $\boldsymbol{x}$ from undersampled measurements $\boldsymbol{b} = \boldsymbol{Ax} + \boldsymbol{n}$.
- **Learned equilibrium**: $\lambda \boldsymbol{A}^H(\boldsymbol{Ax}^* - \boldsymbol{b}) + \boldsymbol{F_\theta}(\boldsymbol{x}^*) = \boldsymbol{0}, \;\; \boldsymbol{F_\theta} = \boldsymbol{I} - \boldsymbol{H_\theta}$ where $\boldsymbol{H_\theta}$ is a learned CNN denoiser.
- **Update**: $\boldsymbol{x}_{k+1} = (\boldsymbol{I} + \alpha\lambda \boldsymbol{A}^H\boldsymbol{A})^{-1}(\boldsymbol{x}_k - \alpha \boldsymbol{F_\theta}(\boldsymbol{x}_k) + \alpha\lambda \boldsymbol{A}^H\boldsymbol{b})$
- **Benefit**: $\boldsymbol{H_\theta}$ is $(1-m)$-Lipschitz for $m \in (0,1) \rightarrow \boldsymbol{F_\theta}$ is m-monotone, $(2-m)$-Lipschitz. Monotonicity supports uniqueness, convergence, and robustness. -
- **Bottleneck**: Each equilibrium iteration requires solving the prox linear system $(\boldsymbol{I} + \alpha\lambda \boldsymbol{A}^H\boldsymbol{A})^{-1}$ iteratively which is expensive if $\boldsymbol{A}^H\boldsymbol{A}$ is large-scale and ill-conditioned.

Proposed: Variable-Metric MOL (VM-MOL)

Let $\boldsymbol{K} = \boldsymbol{A}^H\boldsymbol{A} \in \mathbb{S}_+^n$.

1. Generalized update [2]. : $\boldsymbol{x}_{k+1} = (\boldsymbol{M} + \lambda \boldsymbol{K})^{-1}(\boldsymbol{M}\boldsymbol{x}_k - \boldsymbol{F_\theta}(\boldsymbol{x}_k) + \lambda \boldsymbol{A}^H\boldsymbol{b}), \;\; \boldsymbol{M} \succ \boldsymbol{0}$.
2. Construct the randomized Nyström approximation $\widehat{\boldsymbol{K}} = (\boldsymbol{K\Omega})(\boldsymbol{\Omega}^H\boldsymbol{K\Omega})^\dagger(\boldsymbol{K\Omega})^H = \widehat{\boldsymbol{U}}_\ell \widehat{\boldsymbol{D}}_\ell \widehat{\boldsymbol{U}}_\ell^H$ where $\boldsymbol{\Omega} \in \mathbb{C}^{n \times \ell}$ is a standard normal sketch [3].
3. Choose the deflation metric: $\boldsymbol{M} = \gamma \boldsymbol{I} - \lambda\left(\boldsymbol{K} - \widehat{\boldsymbol{K}}\right) \rightarrow \boldsymbol{M} + \lambda\boldsymbol{K} = \gamma\boldsymbol{I} + \lambda\widehat{\boldsymbol{K}}$. The diagonal + low-rank inverse $\left(\gamma\boldsymbol{I} + \lambda\widehat{\boldsymbol{K}}\right)^{-1}$ can be applied directly in $O(n\ell)$ with the Woodbury identity, avoiding iterative solves per-iteration.

Analysis: High-Probability Tail Bounds in $\mathbb{C}$

1. **Complex Nyström approximation error**: We extend the real-valued Nyström error analysis in [4,5]. Let $\boldsymbol{K} \succcurlyeq \boldsymbol{0}$ have eigenvalues $\mu_1 \geq \cdots \geq \mu_n$. Draw sketch $\boldsymbol{\Omega} \in \mathbb{C}^{n \times (k+p)}$ with iid $\mathcal{CN}(0,1)$ entries where $\mathrm{rank}(\mathbf{K}) \geq k \geq 2$ is the rank and $p \geq 4$ is oversampling. Define:

$$b_p = \frac{e}{p+1}\left(\frac{ep}{\sqrt{2\pi(p+1)}}\right)^{\frac{1}{p}}, \;\; d_1 = b_p k, \;\; d_2 = \frac{e^2(k+p)}{(p+1)^2}.$$

- For $\forall t, u \geq 1$:
  $\Pr\left[\left\|\boldsymbol{K} - \widehat{\boldsymbol{K}}\right\|_2 \leq [1 + 4(d_1 + d_2u^2)t^2]\mu_{k+1} + 4d_2t^2\sum_{j>k}\mu_j\right] \geq 1 - 2t^{-2p} - e^{-u^2}$.
- Interpretation: Error is controlled by first omitted eigenvalue + the remaining spectral mass. Oversampling strengthens the error bound.

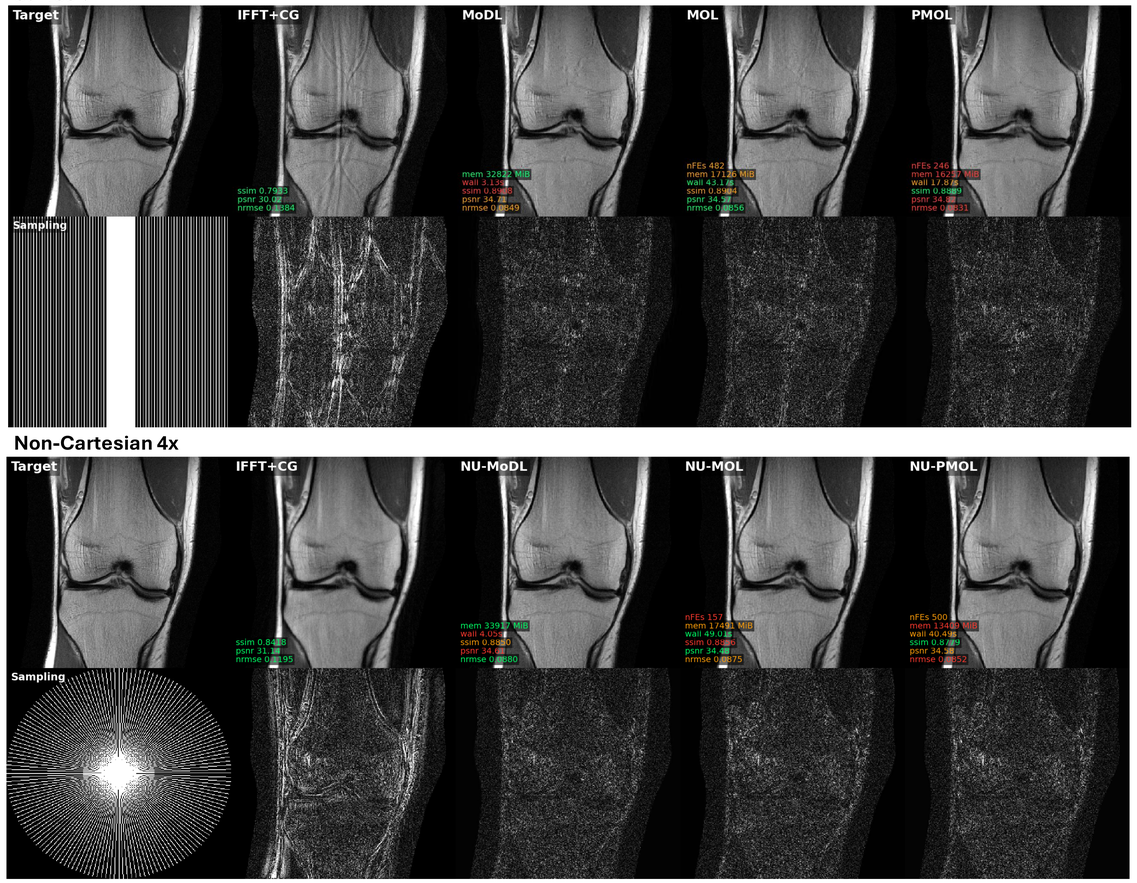
2. **Convergence**: Let $t = \left(\frac{4}{\delta}\right)^{\frac{1}{2p}}, u = \sqrt{\ln\frac{2}{\delta}}$, and B_δ be the resulting error bound. Then:

$$\Pr\left[\left\|\boldsymbol{K} - \widehat{\boldsymbol{K}}\right\|_2 \leq B_\delta\right] \geq 1 - \delta.$$

- On this event, the metric satisfies: $(\gamma - \lambda B_\delta)\boldsymbol{I} \preccurlyeq \boldsymbol{M} \preccurlyeq \gamma\boldsymbol{I}$ which implies $\gamma > \lambda B_\delta$ guarantees $\boldsymbol{M} \succ \boldsymbol{0}$.
- If $\boldsymbol{F_\theta}$ is m-monotone, $(2-m)$-Lipschitz, choosing $\gamma > \max\left\{\lambda B_\delta, \frac{(2-m+\lambda B_\delta)^2}{2m}\right\}$ guarantees contraction.
- The probability is over the random sketch; for a fixed sketch satisfying the error bound, then stability and contraction follow deterministically.

Experimental Results

100 epochs, batch size = 4	MOL	VM-MOL	Speedup
Cartesian	295.86 mins	131.59 mins	2.25x
Non-Cartesian	324.87 mins	229.35 mins	1.42x



(Top) Training wall-time comparison (top) between MOL and VM-MOL. (Middle) Cartesian and (Bottom)non-Cartesian inference reconstruction for IFFT+CG (aka SENSE), MoDL, MOL, and PMOL with no perturbation. Red indicates the best, orange is second best, and green is third or fourth. VM-MOL reduces training time relative to MOL while maintaining comparable reconstruction quality. Reconstructions and error maps are shown for 4× Cartesian and non-Cartesian sampling.

Insights & Next Steps

- Deflation preserves the equilibrium while replacing nested iterative solves with a direct, structured inverse.
- Sketch accuracy and storage determine the practical computational tradeoff.
- **Next**: tighter error bound estimates for selecting γ, rank/oversampling study, application to Kronecker-separable 3D forward operators.

References

[1] Pramanik, Zimmerman, and Jacob, "Memory-efficient model-based deep learning with convergence and robustness guarantees", *IEEE TCI* (2023).
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[3] Martinsson and Tropp, "Randomized numerical linear algebra: foundations & algorithms", *Acta Numerica* (2020).
[4] Persson, Boullé, and Kressner, "Randomized nyström approximation for non-negative self-adjoint operators", *SIAM JMDS* (2025).
[5] Halko, Martinsson, and Tropp, "Finding structure with randomness: probabilistic algorithms for constructing approximate matrix decompositions", *SIAM Review* (2011).

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